

Practice Questions from Week 4

Question 12 : 640653738042

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Assume that you have a dataset of six points $\{x_1, x_2, x_3, x_4, x_5, x_6\}$, all of which are non-negative. You hypothesise that the data points are iid random variables with the following density:

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Discussions (0)



Question 13 : 640653738043

View Parent QN

View Solutions (0)

Total Mark : 2.00 | Type : MCQ

What is the log-likelihood of this dataset under this distribution? $\ln$ represents the natural logarithm or loge.

OPTIONS :

☐ $\prod_{i=1}^6 \lambda e^{-\lambda x_i}$

☐ $\sum_{i=1}^6 \lambda e^{-\lambda x_i}$

☐ $\prod_{i=1}^6 [\ln(\lambda) - \lambda x_i]$

☐ $\sum_{i=1}^6 [\ln(\lambda) - \lambda x_i]$

Solution

$$\mathcal{L}(\theta, \mathcal{D}) = \prod_{i=1}^6 f(x_i; \lambda)$$

$$\mathcal{L}(\theta, \mathcal{D}) = \prod_{i=1}^6 \lambda e^{-\lambda x_i}$$

$$l(\theta, \mathcal{D}) = \ln \left(\prod_{i=1}^6 \lambda e^{-\lambda x_i} \right)$$

$$l(\theta, \mathcal{D}) = \sum_{i=1}^6 \ln(\lambda) + \ln(e^{-\lambda x_i})$$

$$l(\theta, \mathcal{D}) = \sum_{i=1}^6 \ln(\lambda) - \lambda x_i$$

Question 14 : 640653738044[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

You are given the actual values of these observations:

$x_1 = 1, \quad x_2 = 2, \quad x_3 = 3, \quad x_4 = 4, \quad x_5 = 5, \quad x_6 = 6$

What is the maximum likelihood estimate for λ ? Enter your answer correct to three decimal places.

Answer (Numeric):

Answer

Solution

$$\begin{aligned}l(\theta, \mathcal{D}) &= \sum_{i=1}^6 \ln(\lambda) - \lambda x_i \\&= 6 \ln(\lambda) - \lambda \sum_{i=1}^6 x_i \\ \frac{d}{d\lambda} l(\theta, \mathcal{D}) &= \frac{6}{\lambda} - \sum_{i=1}^6 x_i\end{aligned}$$

$$\begin{aligned}\frac{d}{d\lambda} l(\theta, \mathcal{D}) &= 0 \\ \Rightarrow \frac{6}{\lambda} - \sum_{i=1}^6 x_i &= 0 \\ \Rightarrow \frac{6}{\lambda} &= \sum_{i=1}^6 x_i \\ \Rightarrow \hat{\lambda}_{ml} &= \frac{6}{\sum_{i=1}^6 x_i} \\ \Rightarrow \hat{\lambda}_{ml} &= \frac{6}{6+5+4+3+2+1} = \frac{6}{21} \approx 0.285\end{aligned}$$

For the given Beta distribution, choose the correct option. 

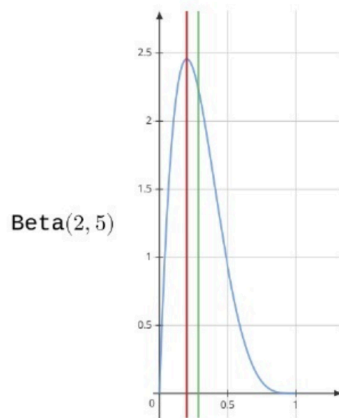


Figure 1: Beta(2,5)

OPTIONS :

☐ Red line denotes the mean for the given Beta distribution, i.e. mean = $\frac{2}{7}$ 

☐ Green line denotes the mean for the given Beta distribution, i.e. mean = $\frac{2}{7}$ 

☐ Red line denotes the mode for the given Beta distribution, i.e. mode = $\frac{1}{5}$ 

☐ Green line denotes the mode for the given Beta distribution, i.e. mode = $\frac{1}{5}$ 

Solution

Given, Beta(2,5) ; $\alpha > 1$, $\beta > 1$

$$\text{Mean of the Beta Distribution} = \frac{\alpha}{\alpha + \beta} = \frac{2}{7} = 0.285 \rightarrow \text{Green Line}$$

$$\text{Mode of the Beta Distribution} = \frac{\alpha - 1}{\alpha + \beta - 2} = \frac{1}{5} = 0.2 \rightarrow \text{Red Line}$$

Question 17 : 640653738041[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : SA

Consider a dataset with 100 total data points. Each data point is classified as either type A or type B. We model this using a Bernoulli distribution, where p is the probability of a data point being type A. If the maximum likelihood estimate (MLE) of p based on the dataset is 0.4, how many data points of type B are there in this dataset?

Answer (Numeric):

Solution

We are given a dataset with 100 total data points

The probability of a data point being Type A is $p = 0.4$

The probability of a data point being Type B is $(1 - p) = (1 - 0.4) = 0.6$

The total number of Type B points is $100 \times 0.6 = 60$

Question 6 : 640653827439[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

The EM algorithm is run on a dataset of 100 points modeled using a GMM with three components. Let λ_k^i be the values obtained in the E-step in the final iteration. Here, i is the index of the data-point and k is the index of a component. Also, let $\pi_1 = 0.3, \pi_2 = 0.5, \pi_3 = 0.2$ be the estimates of the mixture probabilities. x_p represent one of the data-points, for some constant p . If $\lambda_1^p + \lambda_3^p = 0.2$, what is the probability that the second component generated x_p given that we have seen the data-point?



Answer (Numeric):

Solution*Given,*

$$\pi_1 = 0.3, \pi_2 = 0.5, \pi_3 = 0.2$$

$$\lambda_1^p + \lambda_3^p = 0.2$$

we know,

$$\begin{aligned} \sum_{k=1}^K \lambda_k^i &= 1 \\ \implies \lambda_1^p + \lambda_2^p + \lambda_3^p &= 1 \\ \implies \lambda_2^p &= 1 - 0.2 = 0.8 \end{aligned}$$

Question 16 : 640653827450

Total Mark : 0.00 | Type : COMPREHENSION

In the context of Bayesian estimation, consider a Beta prior for the parameter p of a Bernoulli distribution: The dataset has 8 ones and 5 zeros. Based on the above data, answer the given subquestions.

$$p \sim \text{Beta}(4, 3)$$

[Discussions \(0\)](#)**Question 17 : 640653827451**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MCQ

What is the posterior?

OPTIONS :

- ☐ Beta(12, 8)
- ☐ Beta(11, 7)
- ☐ Br(0.5)
- ☐ Beta(8, 5)

Solution

$$\begin{aligned} \text{posterior} &= \text{Beta}(\alpha + n_s, \beta + n_f) \\ &= \text{Beta}(4 + 8, 3 + 5) \\ &= \text{Beta}(12, 8) \end{aligned}$$

Question 18 : 640653827452[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

If we use the expected value of the posterior as a point-estimate for the parameter of the Bernoulli distribution, what is $\hat{p}$?



Answer (Numeric):

Solution

$$E[\text{posterior}] = \frac{\alpha}{\alpha + \beta} = \frac{12}{20} = 0.6$$

Question 6 : 640653992025[View Solutions \(0\)](#) Total Mark : 5.00 | Type : MSQ

Let $X_1, X_2, \dots, X_n$ be n i.i.d. samples with parameter θ , which follows one of the following PDFs:

For $\theta = -1$, we have

$$f(x | \theta) = \begin{cases} 5x^4, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}.$$

For $\theta = 1$, we have

$$f(x | \theta) = \begin{cases} 1, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}.$$

Suppose we wish to find the maximum likelihood estimate of θ , then which among the following are true?

OPTIONS :

☐ If $\prod_{i=1}^n 5x_i^4 < 1$, then $\hat{\theta}_{MLE} = 1$ 

☐ If $\prod_{i=1}^n 5x_i^4 > 1$, then $\hat{\theta}_{MLE} = 1$ 

☐ If $\prod_{i=1}^n 5x_i^4 < 1$, then $\hat{\theta}_{MLE} = -1$ 

☐ If $\prod_{i=1}^n 5x_i^4 > 1$, then $\hat{\theta}_{MLE} = -1$ 

Solution

Given, $X_1, X_2, \dots, X_n \sim iid$ with parameter θ

We have two possible distributions for the data:

1. When $\theta = -1$:

$$f(x|\theta = -1) = \begin{cases} 5x^4, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

2. When $\theta = 1$:

$$f(x|\theta = 1) = \begin{cases} 1, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

→ for $\theta = -1$, Numbers closer to 1 are more likely than numbers closer to 0.

The probability of a number x appearing is $5x^4$, This means that bigger x values (closer to 1) are more likely than smaller x values (closer to 0).

→ for $\theta = 1$, if we pick a number x , its probability is always 1 (*as long as* $0 < x < 1$).

This means every number has the same chance of appearing.

Likelihood for $\theta = 1$,

Since the probability of each number is just 1, the likelihood is:

$$L(\theta = 1) = \prod_{i=1}^n f(x_i | \theta = 1)$$

$$L(\theta = 1) = 1 \times 1 \times \dots \times 1 = 1$$

so for $\theta = 1$, the likelihood is always 1.

Likelihood for $\theta = -1$,

Since the probability for each number is 5 times x^4 , the likelihood is:

$$L(\theta = -1) = \prod_{i=1}^n f(x_i | \theta = -1)$$

$$\begin{aligned} L(\theta = -1) &= 5x_1^4 \times 5x_2^4 \times \dots \times 5x_n^4 \\ &= 5^n \times (x_1^4 \times x_2^4 \times \dots \times x_n^4) \end{aligned}$$

$$= 5^n \prod_{i=1}^n x_i^4$$

so for $\theta = -1$, the likelihood is $5^n \prod_{i=1}^n x_i^4$

If $L(\theta = -1) > L(\theta = 1)$, then $\theta = -1$ is the better choice.

If $L(\theta = -1) < L(\theta = 1)$, then $\theta = 1$ is the better choice.

$$\prod_{i=1}^n 5X_i^4 > 1 \Rightarrow \theta_{MLE} = -1$$

$$\prod_{i=1}^n 5X_i^4 < 1 \Rightarrow \theta_{MLE} = 1$$

Question 9 : 640653992026[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : SA

Consider a GMM for 5 points:

$$x_1 = 1, x_2 = 1.2, x_3 = 2, x_4 = 1.5, x_5 = 0.5$$

At some time-step in the EM algorithm, following are the values of λ_k^i for the k -th mixture after the E-step:

$$\lambda_k^1 = 0.3, \lambda_k^2 = 0.1, \lambda_k^3 = 2.5, \lambda_k^4 = 0.6, \lambda_k^5 = 0.8$$

What is the estimate of μ_k after the M-step? Enter your answer correct to two decimal places.

Answer (Numeric):

Solution

$$\hat{\mu}_k^{mml} = \frac{\sum_{i=1}^n \lambda_k^i \cdot x_i}{\sum_{i=1}^n \lambda_k^i} = \frac{0.3(1) + 0.1(1.2) + 2.5(2) + 0.6(1.5) + 0.8(0.5)}{0.3 + 0.1 + 2.5 + 0.6 + 0.8} = \frac{6.72}{4.3} = 1.56$$

Question 10 : 640653992027[View Solutions \(0\)](#) Total Mark : 4.00 | Type : SA

Consider a dataset with 100 total data points. Each data point is classified as either type A or type B. We model this using a Bernoulli distribution, where p is the probability of a data point being type A. If the maximum likelihood estimate (MLE) of p based on the dataset is 0.4, how many data points of type B are there in this dataset?

Answer (Numeric):

Solution

We are given a dataset with 100 total data points

The probability of a data point being Type A is $p = 0.4$

The probability of a data point being Type B is $(1 - p) = (1 - 0.4) = 0.6$

The total number of Type B points is $100 \times 0.6 = 60$